EXP.NO: 6	Write a Servlet to demonstrate session tracking using HttpSession.

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based web application. The exercise involves implementing a simple login system where a user can log in with their username and password, and their session will be tracked throughout their interaction with the web application.

ALGORITHM:

1. Input:

- The user will enter their **username** and **password** in a **login form**.
- The servlet will validate the credentials and create an **HTTP session** for the user if authentication is successful.

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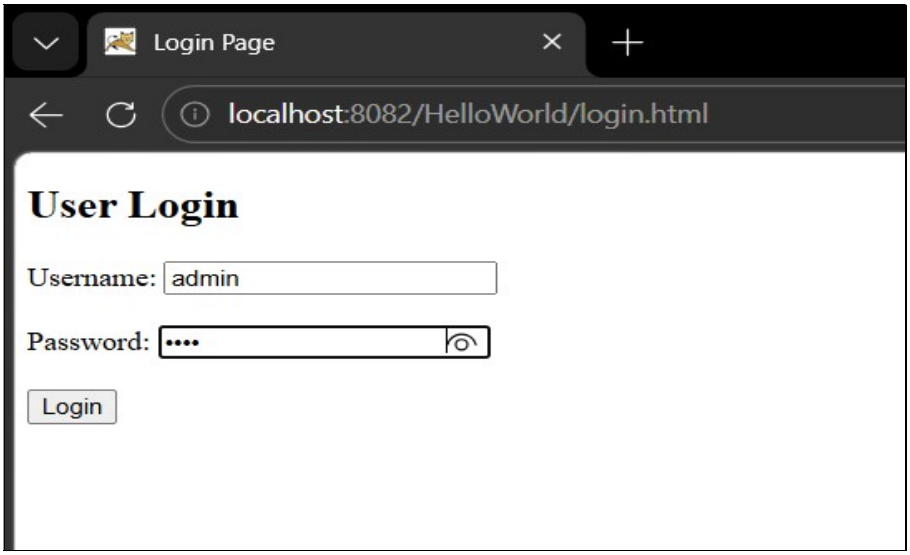
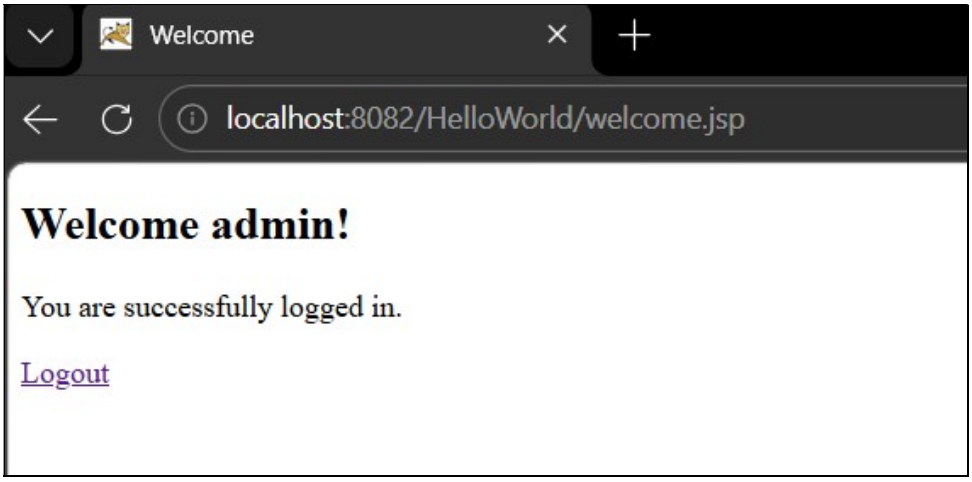
2. Steps for Implementing Session Tracking with HttpSession:

1. **Create the Login Form (HTML):**
   - Create an HTML form with two input fields for **username** and **password**.
   - Provide a **submit button** to send the data to the servlet for authentication.
2. **Create the Login Servlet:**
   - Write a Java servlet class to handle **login** requests.
   - The servlet will:
     - Retrieve the **username** and **password** from the form.
     - Authenticate the user by comparing the credentials to a stored value (e.g., hardcoded or from a database).
     - Create a new **HTTP session** for the user if the credentials are valid.
     - Set session attributes such as the user's **username** to track the session.
3. **Validate User Credentials:**
   - In the servlet, check if the **username** and **password** match the expected values.
   - If valid, create a session and store the **username** in the session.
   - If invalid, redirect the user back to the login page with an error message.

4. **Create and Track User Session:**
    - Upon successful login, use the HttpSession object to store the session data.
    - Use the session.setAttribute("username", username) method to save the **username**.
    - Optionally, set session timeout using session.setMaxInactiveInterval(time_in_seconds) to manage session expiration.
5. **Redirect to a Protected Page:**
    - After a successful login, the user will be redirected to a **protected page** (e.g., a dashboard or home page).
    - The protected page should check if the user is authenticated by checking the **session attribute**.
6. **Logout and Session Invalidation:**
    - Provide a **logout feature** that invalidates the session using session.invalidate() when the user logs out.
    - After logout, redirect the user back to the login page.
7. **Session Expiration:**
    - If the session expires or the user manually logs out, the session will be invalidated, and the user will be redirected to the login page.

Output:





HTML File (login.html):

```
<!DOCTYPE html>
<html>
<head>
    <title>Login Page</title>
</head>
<body>
    <h2>User Login</h2>

    <form action="login" method="post">
        Username: <input type="text" name="username" required><br><br>
        Password: <input type="password" name="password" required><br><br>
        <input type="submit" value="Login">
    </form>

</body>
</html>
```

Java File (LoginServlet.java)

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/login")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        // Hardcoded credentials
        if ("admin".equals(username) && "1234".equals(password)) {

            // Create session
```

```
            HttpSession session = request.getSession();
            session.setAttribute("username", username);

            // Session timeout: 5 minutes
            session.setMaxInactiveInterval(300);

            response.sendRedirect("welcome.jsp");

        } else {
            out.println("<h3 style='color:red'>Invalid Username or Password</h3>");
            request.getRequestDispatcher("login.html").include(request, response);
        }
    }
}
```

Java File (LogoutServlet.java)

```
import java.io.IOException;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/logout")
public class LogoutServlet extends HttpServlet {
```

```
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if (session != null) {
            session.invalidate();
        }

        response.sendRedirect("login.html");
    }
}
```

Java Compiled File (LoginServlet.class) :

// Source code is decompiled from a .class file using FernFlower decompiler (from Intellij IDEA).

```
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
import java.io.IOException;
import java.io.PrintWriter;

@WebServlet({"/login"})
public class LoginServlet extends HttpServlet {
    public LoginServlet() {
    }
```

```
    protected void doPost(HttpServletRequest var1, HttpServletResponse var2) throws ServletException, IOException {
      var2.setContentType("text/html");
      PrintWriter var3 = var2.getWriter();
      String var4 = var1.getParameter("username");
      String var5 = var1.getParameter("password");
      if ("admin".equals(var4) && "1234".equals(var5)) {
        HttpSession var6 = var1.getSession();
        var6.setAttribute("username", var4);
        var6.setMaxInactiveInterval(300);
        var2.sendRedirect("welcome.jsp");
      } else {
        var3.println("<h3 style='color:red'>Invalid Username or Password</h3>");
        var1.getRequestDispatcher("login.html").include(var1, var2);
      }

  }
}
```

Java Compiled File (LogoutServlet.class) :

```
// Source code is decompiled from a .class file using FernFlower decompiler (from Intellij IDEA).
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
```

```
import java.io.IOException;


@WebServlet({"/logout"})
public class LogoutServlet extends HttpServlet {
  public LogoutServlet() {
  }


   protected void doGet(HttpServletRequest var1, HttpServletResponse var2) throws ServletException, IOException {
    HttpSession var3 = var1.getSession(false);
    if (var3 != null) {
      var3.invalidate();
    }


    var2.sendRedirect("login.html");
  }
}
```

File (welcome.jsp) :

```
<%@ page session="true" %>
<%
   String user = (String) session.getAttribute("username");


   if (user == null) {
      response.sendRedirect("login.html");
      return;
   }
%>
```

```
<!DOCTYPE html>
<html>
<head>
    <title>Welcome</title>
</head>
<body>
    <h2>Welcome <%= user %>!</h2>
    <p>You are successfully logged in.</p>

    <a href="logout">Logout</a>
</body>
</html>
```

RESULT:

Thus, A simple login system was created where the user's session is tracked by a servlet to demonstrate session tracking using HttpSession.